Available online at www.sciencedirect.com

ScienceDirect

Aquaculture 273 (2007) 711 – 720

Aquaculture

www.elsevier.com/locate/aqua-online

Effects of salinity and pH on ion-transport enzyme activities, survival and growth of *Litopenaeus vannamei* postlarvae

Lu-Qing Pan *, Lin-Juan Zhang, Hong-Yu Liu

The key laboratory of Mariculture, Ministry of Education, Ocean University of China, Qingdao 266003, China

Received 14 November 2006; received in revised form 17 July 2007; accepted 17 July 2007

Abstract

Effects of the salinity and pH on ion-transport enzyme activities, survival and growth of *Litopenaeus vannamei* postlarvae were investigated. Shrimp were transferred from salinity 31‰ and pH 8.1 to different salinity levels of 22, 25, 28, 31 (control), and to different pH levels of 7.1, 7.6, 8.1 (control), 8.6 and 9.1. The results showed ion-transport enzyme activities and weight gains of postlarvae were significantly affected by salinity and pH variation, which had no obvious effect on survival rate. The changing salinity affected the activities of total ATPase and Na^+–K^+-ATPase notably ($F > F_{0.05}$), meanwhile, non-significantly to the activities of V-ATPase, HCO_3^--ATPase. Within 48 h of salinity changing, the activities of ATPase, Na^+–K^+-ATPase in each treatment group gradually increased with the sampling time and reached their climax at 48 h, and then stabilized, showing negative correlation with salinity. The changing of pH affected greatly the activities of ATPase, Na^+–K^+-ATPase, V-ATPase and HCO_3^--ATPase ($F > F_{0.05}$), the activities of ATPase, Na^+–K^+-ATPase in each treatment group (pH = 7.1, 7.6, 8.6, 9.1) showed peak change within 72 h and stabilized afterwards, and the Na^+–K^+-ATPase activities came back to the level of the control group; Meanwhile the changing extent of V-ATPase and HCO_3^--ATPase activity corresponded with the grads of pH, and these ATPase activities showed negative correlation with pH changing, the activities of V-ATPase, HCO_3^--ATPase in postlarvae of each treatment group came to stable level after 24 h. The experiment also indicated the strength order of these ion-transport enzyme activities were as follows: Na^+–K^+-ATPase > V-ATPase > HCO_3^--ATPase. Na^+–K^+-ATPase was the chief undertaker of osmoregulation under the salinity effects, while V-ATPase and HCO_3^--ATPase were the chief osmoregulation undertakers under pH changing. In different salinity environment, the contributions of Na^+–K^+-ATPase, V-ATPase and HCO_3^--ATPase of *L. vannamei* postlarvae approximately accounted for 62.0–78.0%, 15.9–29.0% and 2.03–4.12% of ATPase activities in total, respectively. Meanwhile, in different pH medium, the contributions of these ATPases approximately accounted for 50.7–67.4%, 21.3–31.8% and 2.15–7.90% of total ATPase activities, respectively. Weight gain of shrimp transferred to salinity 31 (control) and 28‰ was significantly higher than that of shrimp reared at 25 and 22‰, and weight gain of shrimp transferred to pH 8.1 (control) and 8.6 was significantly higher that that of shrimp transferred to pH 7.1, 7.6 and 9.1. It was suggested that during the process of desalting and culturing of postlarvae, the salinity changing should not exceed 3 and pH variety not more than 0.5.

Keywords: Salinity; pH; *Litopenaeus vannamei* postlarvae; Ion-transport enzyme activities; Survival; Growth

1. Introduction

Litopenaeus vannamei has got high commercial importance, excellent breeding property as well as strong

* Corresponding author. Tel.: +86 532 82032963; fax: +86 532 82894024.

E-mail address: panlq@ouc.edu.cn (L.-Q. Pan).

0044-8486/$ - see front matter © 2007 Elsevier B.V. All rights reserved.
doi:10.1016/j.aquaculture.2007.07.218

antiviral ability. Because of its euryhaline characteristic, *L. vannamei* has become the main aquatic animals cultured in brackish water as well as freshwater regions in recent years. Due to the various salinities of its cultural water, ruderal and lichen are likely to grow massively, which results in high pH level. Therefore, one salient problem of shrimp farming is adaptability to hypohaline water or freshwater, then its survival and growth; another problem is the regulating standard of its cultural water. Both are relevant directly to its osmoregulation. However, the research of its physiological adaptation is always neglected.

The salinity and pH are affected by many factors. Firstly, excessive freshwater or rainstorm might rapidly make some change; Secondly, the sudden change or disappearance of dominant species of plankton as well as water pollution might cause rapid change of salinity and pH; Thirdly, high evaporation and natural episodic fluxes of freshwater input into estuaries from meteorological events are common and often result in acute salinity fluctuations in estuarine salina. (Christensen et al., 1997; Pan et al., 2006). Therefore, during the process of desalting and culturing shrimps, the variation of salinity and pH leads to the declination of survival rate or slow growth, or even serious diseases and mass death.

In general, crustacean has the ability to adapt to the variety of salinity or pH, as well as other environmental factors. They can keep their normal life functions largely by hemolymph osmoregulation. Regulation of hemolymph osmotic pressure in crustacean mainly depends on inorganic ion (such as Na^+, K^+ and Cl^-) and the concentrations of free amino acid. The concentrations of Na^+ and Cl^- are the most important contributors in hemolymph osmoregulation, which account for about 80–90% of hemolymph osmolality. Many research papers euryhaline species have been published, including *Eriocheir sinensis* (Onken and Putzenlechner, 1995), *Penaeus chinensis* (Chen and Jun, 1994), *Scylla serata* (Chen and Chia, 1997), and *Macrobrachium rosenbergii* (Wilder et al., 1998). Ion-regulation in crustacean is mostly accomplished by Na^+–K^+-ATPase, V-ATPase, HCO_3^--ATPase, Carbonic anhydrase (CA) and many other ion-transport enzymes in gill epithelium membrane (Morris, 2001), among which, the Na^+–K^+-ATPase in freshwater shrimp *Macrobrachium olfersii* accounts for about 70% of the total ATPase activity; the remaining 30% apparently correspond to V- and F_0F_1-ATPases (Furriel et al., 2000).

The osmoregulatory capacity measurement was confirmed as a convenient tool to monitor the physiological condition and the effect of stress in crustaceans (Charmantier and Soyez, 1994). It has been suggested that branchial Na^+–K^+-ATPase is of central importance in ion regulation and cellular water balance by aquatic vertebrates and invertebrates, and directly related to their osmoregulation capacity (Towle, 1981). The activity of Na^+–K^+-ATPase reflected the osmoregulation of crustaceans. The Na^+–K^+-ATPase was positioned to pump sodium ions from the cytosol across the basolateral membrane into the extracellular fluid in exchange for potassium ions moving in the opposite direction (reviewed by Lucu and Towle, 2003). When euryhaline crustaceans are transferred from natural seawater to dilute seawater, their Na^+–K^+-ATPase activities increase markedly, vice versa. Specific activity in gills of many strongly hyperosmoregulating species has been reported, including *Penaeus monodon* (Ferraris et al., 1986), *M. olfersii* (Lima et al., 1997; Furriel et al., 2000), *L. vannamei* (Davis et al., 2002; Saoud et al., 2003), *Penaeus stylirostris* and *Penaeus setiferus* (Castille and Lawrence, 1981), and *Penaeus chinensis* (Cheng and Chen, 1998).

V-ATPase and HCO_3^--ATPase are located in the apical membranes of the epithelial cells of gills. Outward pumping of H^+ by V-ATPase would generate a large electrical potential across the apical membrane and provide an inwardly directed electrochemical gradient for Na^+ driving entry of Na^+ via sodium channels in the apical membrane (Morris, 2001; Onken and Putzenlechner, 1998), and provide the driving force for coupled Cl^-/HCO_3^- exchanger; HCO_3^--ATPase may be involved in Cl^- uptake, and related to Cl^-/HCO_3^- exchange (Lee, 1982). Zare and Greenaway (1998) observed that Sodium depletion in *Cherax destructor* caused V-ATPase to be Na^+–K^+-ATPase co-operating helper to participate ion-transport by Na^+/H^+ exchanger; Forgac (1998) had studied the structure of V-ATPase, and concluded that V-ATPase can take part in osmoregulation by modulating the concentration of H^+ when pH changed. Few research about HCO_3^--ATPase were reported, Lee (1982) found that physiological characterization of gill HCO_3^--ATPase was affected significantly by low salinity, it was possible that the variation were attributed to the enzyme synthesis induced by environmental activation, neither by alteration of membrane structure nor the enhancement of the pre-existing ATPase activity.

The changes of ion-transport enzyme activities in crustacean are related to the external environment, and reflected the osmotic physiological adaptation. So far, there is no study on the effects of salinity and pH on the osmotic physiological mechanism in shrimp postlarvae. The objective of the present research was to evaluate ion-transport enzyme activities (total ATPase, Na^+–K^+-ATPase, V-ATPase and HCO_3^--ATPase), survival and growth of *L. vannamei* postlarvae acclimated to various salinity and pH, and to explore the mechanism of osmotic physiological adaptation in shrimp. Furthermore, some appropriate parameters of osmotic physiological adaptation for the culturing regulation were searched, and we will provide a scientific basis of cultural water regulation for *L. vannamei* farming in brackish water and freshwater.

2. Materials and methods

2.1. Materials

The postlarvae of the shrimp, *L. vannamei*, were obtained from commercial farms in Yinghai, Qingdao. The average biologic length of postlarvae was about 0.8 cm, and their age was 18–20 days. The postlarvae were reared in tanks containing aerated water (salinity 31‰, pH 8.5; temperature 24±0.5 °C) with air-lift, for 3 to5 days prior to experimentation in laboratory. Half of the water in each tank was renewed twice daily. They were fed with commercial prawn pellets twice daily.

2.2. Grads designing of salinity and pH

The experiment consisted of salinity and pH sections, respectively. After acclimation to seawater (salinity 31, pH 8.1), postlarvae were directly placed into each grads medium of salinity and pH separately. The salinity grads of 28, 25, and 22 were designed by diluted seawater with insolated tap water, and pH grads of 7.1, 7.6, 8.6, and 9.1 were set by regulated seawater with 1.0 mol/L NaOH and 1.0 mol/L HCl and varied about ±0.1. In both of the experiment, the natural seawater groups served as control, and there were three replicates of each treatment, meanwhile, the experimental conditions were the same as those used for acclimation. Each experiment was subdivided into ion-transport enzyme activities and growth sections.

Specimens of enzyme activities test were cultured in plastic tanks (50 cm × 40 cm × 30 cm), about 150–200 postlarvae per tank, and there were three replicates of each treatment. Ten specimens were sampled every time at 0, 6, 12, 24, 36, 48, 60, 72, 96 h from each tank, and stored in 1.5 ml eppendorf at −80 °C before assayed.

The experiments of survival and growth were conducted in artificial temperature-controlling tanks. Each container, holding 10 postlarvae, was filled with 500 ml experimental water without aerator. The experiment lasted for 96 h. At the end of the experiment, survivals were counted and daily weight gain was calculated. Three to five postlarvae were taken separately from each tank at the beginning and the end of the experiment. The specimens were rinsed three times in distilled water, then dried at 60 °C and weighted using a microbalance with 0.1 mg precision until the weight was constant. The value of daily body weight gain was based on the following equation:

$$\text{Daily body weight gain}(\text{g/ind} \cdot \text{d}) = \frac{\text{final weight} - \text{initial weight}}{\text{experimental time} \times \text{number of postlarvae}}$$

2.3. Determination of ion-transport enzyme activities

2.3.1. Preparation of enzymes

The samples (about 1.5 g) were homogenized in 4 ml ice-cold homogenization (containing 250 mM Sorbitol, 6 mM EDTA, 25 mM tris/acetate, 0.1 mM DTT, 0.2 mM PMSF, 100U/ml Aprotinin, pH 7.4) at 10,000 r min^{-1} for 5 min. Then Homogenates were centrifuged at 11,000 r min^{-1} for 30 min at 0 °C, and the supernatant medium were centrifuged again at the same condition for 10 min. The final supernatant was stored in a refrigerator at 0–4 °C. The total ATPase, Na^+–K^+-ATPase, V-ATPase and HCO_3^--ATPase activities were measured in 8 h.

2.3.2. Enzyme assay

ATPase activities were measured according to the methods of Lee (1982) and Zare and Greenaway (1998), by measuring the quantity difference of inorganic phosphorus (Pi) released by ATP at the condition of activators existing or not (using inhibitor). The enzyme activity was expressed as μmol P released per mg protein per h.

2.3.2.1. Measurement of Na^+–K^+-ATPase activity. The Enzyme sample (50 μl) was incubated separately in two reaction media (1.1 ml). Buffer (A), containing 6 mM $MgCl_2$, 100 mM NaCl, 10 mM KCl, 25 mM Tris/acetate buffer, was adjusted to pH 7.4, and buffer (B), containing 6 mM $MgCl_2$, 110 mM NaCl, 5 mM ouabain (inhibit Na^+–K^+-ATPase), 25 mM Tris/acetate buffer, was adjusted to pH 7.4. All tubes were equilibrated to 30 °C for 10 min, using purified water (50 μl) served as controls. The reaction was initiated by adding 110 μl of 10 mM disodium ATP to each tube, which provided an initial ATP concentration of approximately 1 mM. After 15 min the reactions were stopped by the addition of 300 μl TCA (30%). The supernatants were assayed for inorganic phosphate following ammonium molybdate ascorbic acid method. The total ATPase activity was determined by the quantity of inorganic phosphate in buffer (A). Protein concentration was determined by the method of Bradford (1976).

2.3.2.2. Measurement of V-ATPase activity. The assay buffer contained 34.8 mM Hepes, 173.9 mM KCl, 6.95 mM $MgCl_2$, 0.01 mM orthovanadate, pH 7.0; the orthovanadate was dissolved in deionised water, then adjusted to pH 7.4 with HCl (0.5 mM) and then boiled for 2 min to destroy polyxyanions before adding to the assay buffer. In addition to this buffer, a second solution of similar composition was prepared, which contained the V-ATPase inhibitor Bafilomycin A_1, to give an assay concentration of 16 μM. Assay buffer (847 μl) and sample homogenate (37 μl) were mixed in tubes and equilibrated to 30 °C. DMSO (16 μl) was added to the “non-inhibited” assays while DMSO plusing bafilomycin A_1 (16 μl) was added to the “inhibited” assays. The reaction was started by the addition of 100 μl of disodium ATP (30 mM) to each tube. After exactly 10 min, 150 µl TCA (40%) was added to stop the reaction, and the supernatant was assayed for Pi, as described above.

2.3.2.3. Measurement of HCO_3^--ATPase activity. The buffer (A) contained 50 mM $NaHCO_3$, 3 mM $MgCl_2$, 50 mM Tris–Hepes buffer (pH 7.8); buffer (B), composed of 50 mM NaCl, 100 mM NaSCN, 3 mM $MgCl_2$, 50 mM Tris–Hepes buffer (pH 7.8). The Enzyme sample (50 µl) was incubated separately in two reaction media (0.85 ml) and equilibrated to 30 °C for 10 min. All tubes were added 100 µl 10 mM ATP for 15 min (the period of linear activity), then the reaction was terminated by the addition of 1.0 ml TCA (10%) and the supernatant were assayed for Pi, as described above.

2.4. Data analysis

All values were presented as means ± SD (n=3). Statistical methods of one-way ANOVA and F check were used to test the significance of differences among treatments and various time intervals.

3. Results

3.1. Effects of salinity on ion-transport enzyme activities of *L. vannamei* postlarvae

The effects of salinity on total ATPase and Na^+–K^+-ATPase activities of *L. vannamei* postlarvae were significant ($F > F_{0.05}$),

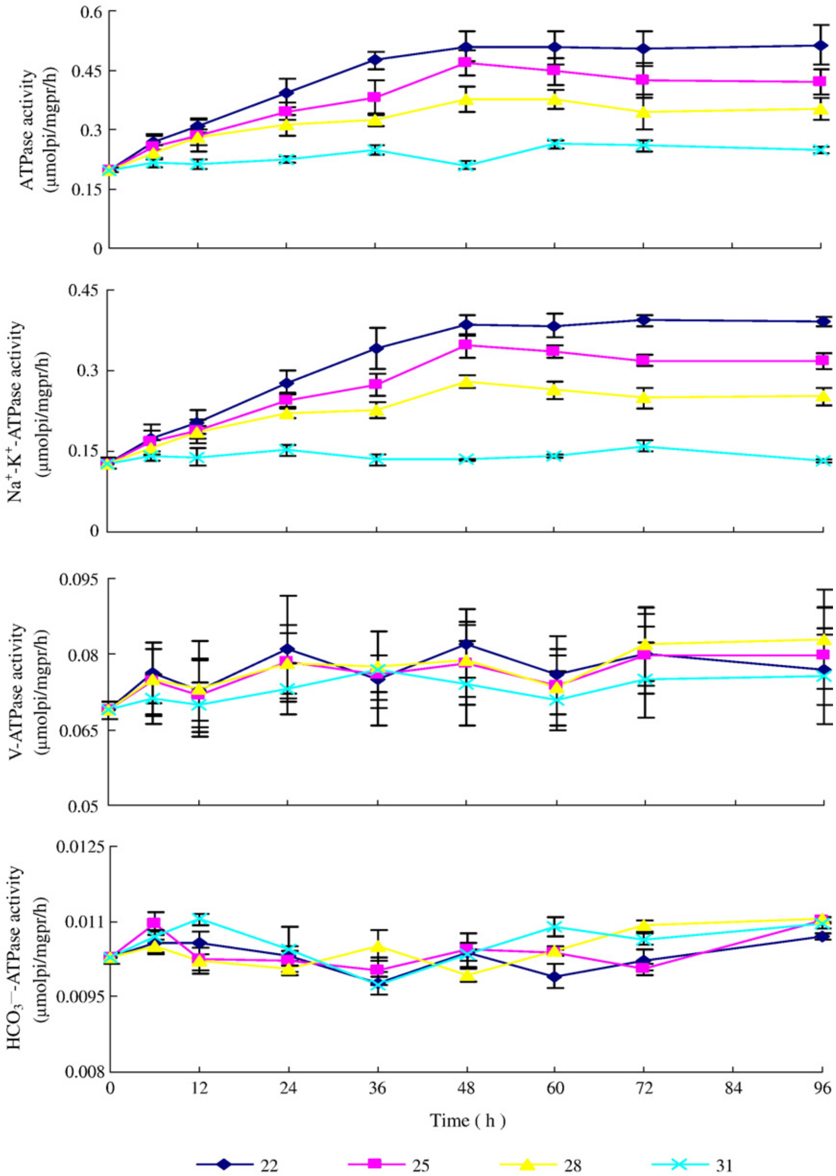

Fig. 1. Effects of salinity on ion-transport enzyme activities of *Litopenaeus vannamei* postlarvae. Values are means ± S.D (n=3). ($F > F_{0.05}$).

but the activities of V-ATPase and HCO_3^--ATPase were unaffected ($F<F_{0.05}$). The activities of total ATPase and Na^+–K^+-ATPase in the experimental groups increased distinctly in 48 h, stabilized afterwards; there was also a negative relation between the levels of the two enzyme activities and salinity (Fig. 1).

3.2. Effects of pH on ion-transport enzyme activities of L. vannamei postlarvae

The ion-transport enzyme activities of *L. vannamei* postlarvae were significantly affected by pH changing ($F>F_{0.05}$;

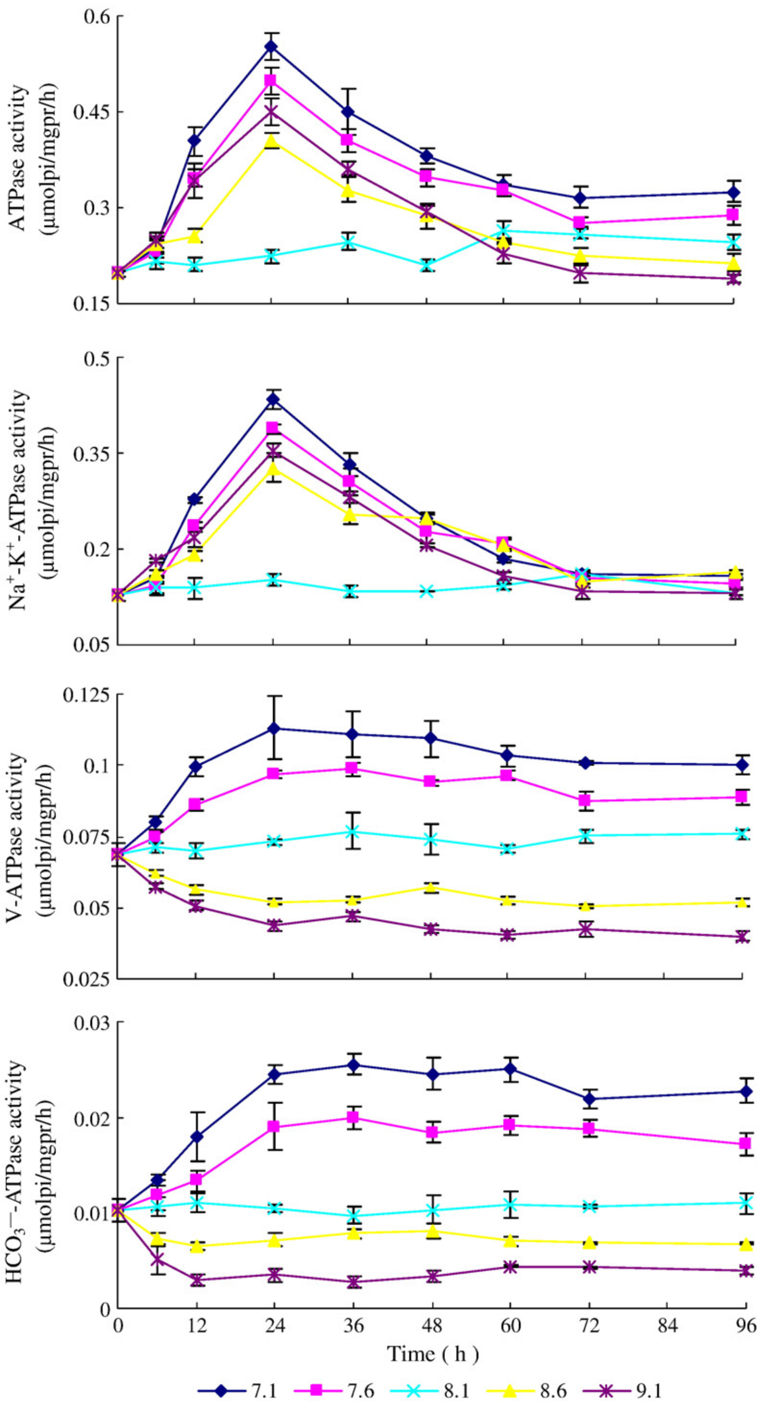

Fig. 2. Effects of pH on ion-transport enzyme activities of *Litopenaeus vannamei* postlarvae. Values are means ± S.D ($n=3$). ($F>F_{0.05}$).

Table 1
Effects of salinity and pH on the contribution of each ion-transport enzyme to the total ATPase activity in *Litopenaeus vannamei* postlarvae

Factors		Total ATPase	Na^+–K^+-ATPase	V-ATPase	HCO_3^--ATPase	Remains of ATPase
Salinity	22	100	78.0	15.9	2.03	4.07
	25	100	75.2	18.8	2.37	3.63
	28	100	72.3	23.8	3.16	0.74
	31	100	62.0	29.0	4.12	4.88
pH	7.1	100	50.7	31.8	7.90	9.60
	7.6	100	56.1	31.5	6.92	5.48
	8.1	100	62.0	29.0	4.12	4.88
	8.6	100	66.8	22.6	3.07	7.53
	9.1	100	67.4	21.3	2.15	9.15

Notes:
In different salinity and pH, the data of ion-transport enzyme activities of *Litopenaeus vannamei* postlarvae were selected at 72 h. We regarded total ATPase activity in each particular grad of salinity (or pH) as 100 percent, respectively, and then we calculated the proportions of Na^+–K^+-ATPase, V-ATPase and HCO_3^--ATPase accounted for total ATPase activities.

Fig. 2). The activities of total ATPase and Na^+–K^+-ATPase showed peak changes in the experimental groups (pH = 7.1, 7.6, 8.6, 9.1) within 72 h, reaching maximum at 24 h, and stabilized after then. The total ATPase activities had significant difference compared with control group (pH = 8.1), and there was a negative correlation with pH, while the Na^+–K^+-ATPase activity showed no significant difference compared with the controls after 72 h. Meanwhile the changing extent of V-ATPase and HCO_3^--ATPase activity corresponded with the grads of pH, and these ATPase activities showed negative correlation with pH changing, the activities of V-ATPase, HCO_3^--ATPase in postlarvae of each treatment group came to stable level after 24 h.

3.3. The contributions of ion-transporting enzyme activities in L. vannamei postlarvae under different salinity and pH

From Table 1, we can see the contributions of Na^+–K^+-ATPase, V-ATPase and HCO_3^--ATPase of *L. vannamei* postlarvae that accounted for total ATPase activity were 62.0–78.0%, 15.9–29.0% and 2.03–4.12%, respectively, in the changes of different salinity. Functioning as the osmoregulation undertaker of postlarvae, the contribution of Na^+–K^+-ATPase increased with the decreasing of salinity. Meanwhile, the contributions of three ion-transporting enzyme activities that accounted for total ATPase activity were 50.7–67.4%, 21.3–31.8% and 2.15–7.90% under different pH grads, respectively, the contributions of V-ATPase and HCO_3^--ATPase of osmoregulation undertaker decreased with pH rising.

3.4. Effects of salinity and pH on survival rate and weight gain of L. vannamei postlarvae

There were no significant effects on the survival rates of *L. vannamei* postlarvae in the variation of salinity and pH, Weight gain of shrimp transferred to salinity 31 (control) and 28‰ was significantly higher than that of shrimp reared at 25 and 22‰, and weight gain of shrimp transferred to pH 8.1 (control) and 8.6 was significantly higher that that of shrimp transferred to pH 7.1, 7.6 and 9.1(Fig. 3).

4. Discussion

4.1. The physiological regulation of the ion-transport enzyme activities during acclimation to salinity and pH in L. vannamei postlarvae

Many scholars had studied the effects of salinity on the branchial ion-transport enzyme activities in crustacean. In the field of the Na^+–K^+-ATPase, Lima et al. (1997) observed that *M. olfersii* gill Na^+–K^+-ATPase activity decreased by ≈35% after acclimation to media of 21‰ and 28‰ salinity for 20 d; López Mañanes et al. (2002) found that *Cyrtograpsus angulatus* were transferred from 35‰ to 6‰ salinity for 3 d, its Na^+–K^+-ATPase activity of posterior gills increased notably in the first day and stabilized afterwards; Neufeld et al. (1980) reported that the activity of Na^+–K^+-ATPase in *Callinectes sapidus* reached stabilization after 8–12 d of transferred to 20‰ salinity from 100‰ seawater, and the activity of low

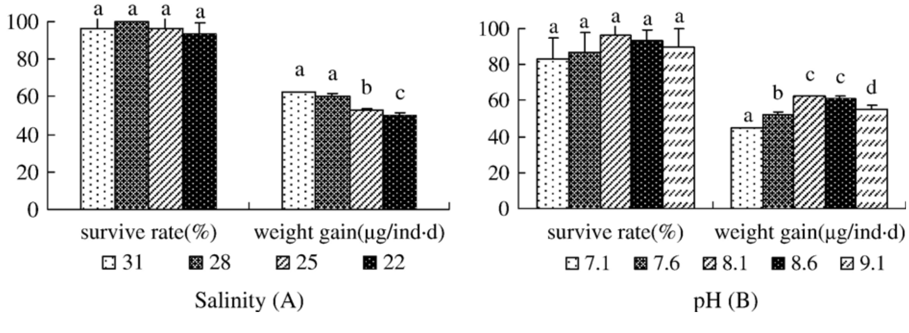

Fig. 3. Effects of salinity and pH on survival rate and weight gain of *L. vannamei* postlarvae. Values are means ± S.D (n = 3). Different letters indicate significant differences ($F > F_{0.05}$).

salinity was stronger than high salinity. In the field of V-ATPase and HCO_3^--ATPase research, there was no significant effect on the V-ATPase activity of *Carcinus maenas* at different salinity levels (Weihrauch et al., 2002); meanwhile, Lee (1982) pointed that the HCO_3^--ATPase activity of gills increased when *C. sapidus* were shifted from seawater (1000 mOsm) to low salinity (100 and 200 mOsm), and stabilized after 14 d. The present results showed that the Na^+–K^+-ATPase activity in each treatment group(S=22, 25, 28) was gradually increased within 48 h of salinity changes and stabilized afterwards, having negative correlation with salinity; While the activities of V-ATPase and HCO_3^--ATPase had no significant difference during experimental time. It was demonstrated that Na^+–K^+-ATPase was the chief undertaker of ionic balance of *L. vannamei* postlarvae, and had obvious temporal regularity during acclimation to salinity. The conclusion was consistent with previous reports, which suggested that gill (Na^+,K^+)-ATPase activity increased in hyperosmoregulating crustaceans, subject to reduced salinity, and decreased in those exposed to elevated salinities (Péqueux, 1995; Lucu and Towle, 2003). In our research, V-ATPase activity had no obvious change during osmoregulatory process, although in freshwater crustaceans, an apical V-ATPase might act in series with the (Na^+,K^+)-ATPase to drive Na^+ uptake from the very dilute external medium (Onken and Riestenpatt, 1998; Zare and Greenaway, 1998; Kirschner, 2004). In additional, the variation of the HCO_3^--ATPase activity did not match Lee's research; we think that different contributions of ion-transport enzymes are decided by different salt tolerance of the two species. Another reason might be the different enzymes activity measurements. The diversity of osmoregulation mechanism in crustacean might contribute to the reason for species difference.

Few researches could be found about the effects of external pH on the ion-transport enzyme activities in crustacean. Shaw (1960) approved that the capacity of absorbing Na^+ in *Astacus pallipes* were inhibited when the concentrations of H^+ outside were at high level. Meanwhile, the decreasing ability of excreting H^+ in gill epithelial cell could led to accumulation of mass H^+ influx. Thus, the increment of V-ATPase and HCO_3^--ATPase promoted the excretion of H^+ and the balance of inside–outside medium. However, publications on optimal pH of gill ion-transport enzyme in crustacean had already been found. It was proved that the optimal pH of gills Na^+–K^+-ATPase in many crustacea was 7.0–8.0, such as *Macrobrachian rosenbergii* (Stem et al., 1984), *Orconectes limosus* (Kosilo et al., 1988), *Hemigrapsus nudus* (Corotto and Holliday, 1996). Breitwieser et al. (1987) found that Na^+–K^+-ATPase of squid was relying on internal pH, whose climax was at pH=7, and that was hardly affected by external pH; Lee (1982) suggested that optimal pH of HCO_3^--ATPase in blue crab was 7.8, that the absorption of Cl^- was added by the activation of HCO_3^--ATPase activity, which also helped Cl^-/ HCO_3^- exchange; In addition, Oliveira et al. (2004) showed that the optimal pH of V-ATPase in *Anodonta cygnea* was 7.0–7.4, the activity decreased markedly when pH>8.0.

Our results indicated that the activities of Na^+–K^+-ATPase in experimental group showed peak changes within 72 h of pH variation and dropped to the level of the control group afterwards; while the activities of V-ATPase and HCO_3^--ATPase stabilized after 24 h, showing negative correlation with pH. Although the Na^+–K^+-ATPase took part in the course of osmoregulation, V-ATPase and HCO_3^--ATPase in postlarvae were the most important undertakers of regulating acid–base equilibrium in vivo, which showed obvious temporal regularity during adaptation process to pH, independent of pH variation. The external pH is predominantly of ecophysiological importance or a tool for the characterization of transport proteins (Onken and Putzenlechner, 1998), and stimulation of Cl^- uptake by ATP was dependent on the pH of the medium (Gerencser and Zhang, 2001). We deem that the changing pH stimulated the activities of V-ATPase (Na^+/H^+) and HCO_3^--ATPase (Cl^-/HCO_3^-), because the functions of ion-transport enzymes were regulated by pH medium, the optimal pH in vitro was about 7–8.

The results also indicated that the strength order of ion-transporting enzyme activities in *L. vannamei* postlarvae, which could be showed: Na^+–K^+-ATPase>V-ATPase>HCO_3^--ATPase. Meanwhile, Na^+–K^+-ATPase of postlarvae was the chief undertaker of osmoregulation under the variety of salinity, while V-ATPase and HCO_3^--ATPase of postlarvae were in charge of osmoregulation in changes of pH. Therefore, in different salinity and pH medium, the balance of osmoregulation in *L. vannamei* postlarvae depended mostly on different ion-transport enzymes, and the acclimatization of ion-transport enzymes fully reflected the physiological adaptation of shrimp osmoregulation. We suggested that Na^+–K^+-ATPase, V-ATPase and HCO_3^--ATPase could be regarded as the suitable parameters of estimating the osmotic physiological adaptation of prawn postlarvae, respectively.

4.2. The physiological mechanism of osmoregulation in L. vannamei postlarvae at different salinity and pH

Wilder et al. (1998) considered the hemolymph osmotic pressure of *Macrobrachium rosenbergii* that mostly depended on the concentration of Na^+, which had proved that Na^+–K^+-ATPase played the most important role in Na^+ absorption; Furriel et al. (2000) suggested that the Na^+–K^+-ATPase of *M. olfersii* corresponded to about 70% of the total ATPase activity; the remaining 30%, i.e. the ouabain-insensitive ATPase activity, apparently corresponded to F_0F_1- and V-ATPases, but not Ca-stimulated and Na- or K-stimulated ATPases. The present research showed that three kinds of ion-transport enzymes of *L. vannamei* postlarvae had obvious variation regularity at different salinity and pH. In different salinity environment, the contributions of Na^+–K^+-ATPase, V-ATPase and HCO_3^--ATPase of postlarvae approximately accounted for 62.0–78.0%, 15.9–29.0% and 2.03–4.12% of ATPase activities in total, respectively. Meanwhile, in different pH medium, the contributions of these ion-transport enzymes accounted for about 50.7–67.4%, 21.3–31.8% and 2.15–7.90% of total ATPase activities, respectively. Those above were corresponded with previous studies in the main. The results showed that the contributions of three kinds of ion-transport enzymes in *L. vannamei* postlarvae were similar in different salinity and pH, or slightly distinctive, which mainly related to the sorts of ion-transport enzymes that maintaining the osmoregulation balance of postlarvae. The results also presented the changing processes and speciality of postlarvae osmoregulation.

There were possible osmotic mechanisms of crustacea. *C. maenas* (Siebers et al., 1982; Lucu and Flik, 1999) pointed out that when transferred from seawater (SW) to diluted SW, Na^+–K^+-ATPase activity in crabs may be regulated by the preexisting enzyme or the recruitment of silent enzyme in the short run, and by producing the new enzyme and/or by reducing the degradation rate in the long-term acclimation. In *C. sapidus*, only the enzyme activity, but not the mRNA or the protein level, showed a significant difference between salinity treatments (Piller et al., 1995; Towle et al., 2001). Long-term increases in Na^+–K^+-ATPase activity might also derive from cellular differentiation processes, resulting in an increased abundance of ion-transporting cells after transferal to dilute media. Alterations in the rates of enzyme synthesis and degradation may also be involved, together with posttranslational processes like subunit assembly, membrane trafficking and cell signaling (Towle et al., 2001; Lucu and Towle, 2003). In changing salinity, we found the variation of bioamine would stimulate pre-existing Na^+–K^+-ATPase activities in short-term adaptation, which regulated ionic concentration of shrimp (in vivo and vitro) rapidly. Therefore, shrimp could maintain hemolymph osmotic pressure stabilization and osmoregulatory balance while holding the natural physiological metabolism; then neuroendocrine system regulation then would affect the enzyme activities by increasing the synthetic locations of enzyme and stimulating new enzymes syntheses. In changing pH, the variation of concentration in H^+/OH^- had stimulated V-ATPase activity, affecting Na^+/H^+ exchange, while Na^+–K^+-ATPase presented emergency reaction in the short-run and came back to normal levels afterwards. Meanwhile, V-ATPase would drive Cl^- uptake and provide the driving force for Cl^-/HCO_3^- exchanger, which might activated HCO_3^--ATPase activity. The activities of V-ATPase and HCO_3^--ATPase stabilized after pH adaptation.

4.3. Effects of salinity and pH on survival rate and growth of L. vannamei postlarvae

There have been many investigations about the effect of salinity on the survival and growth of *Penaeus chinensis*, *L. vannamei* and *Penaeus semisulcatus* juveniles (PZ_1–PL_{18}), the survival rate and weight gain were kept at high level during 40–50 d of the low salinity (10–25), but decreased sharply in high salinity (>40) (Chen and Liu, 1996; Jesus et al., 1997; Kumlu and Eroldogan, 2000); Within 96 h of salinity changes (15–35), the survival rate of juveniles (about 22±2.4 mm) of *L. vannamei* was 100% (Chen and Lin, 2001); Allan and Maguire (1992) reported that-when *Penaeus monodon* (4.2–5.5 g) were cultured in pH at 6.1–7.8, juveniles mostly survived in 96 h, but largely died at pH < 5.1 within 48 h, while weight gain had no significant difference when juveniles were acclimated to pH (7.8, 7.3, 6.7, 6.1) and salinity (15,30) for 23 d; moreover, Wang et al. (2002) found that the survival rate of *P. chinensis* juveniles (20–30 mm) was impaired at pH ≥ 8.5 or pH ≤ 6.0 in 14 d. Our research showed that there were no significant effects on the survival rates o of *L. vannamei* postlarvae at different salinity and pH levels, while weight gain were distinctly affected in experimental groups of S=22, 25; pH=7.1, 7.6, 9.1. The result didn't show a complete homology with previous researches. It proved that *L. vannamei* juveniles have the capacity of adjusting themselves to changing salinity and pH, and the alterant extent of experiment factors were in the range of its osmoregulation. Shrimp might exist two distinct physiological adaptive mechanisms, including short-time and long-time modulation. Rosas and Sanchez (1996) suggested that *Litopenaeus setiferus* required extra energy to regulate the osmotic pressure and ionic concentration in vivo to maintain homeostasis when environmental salinity changed; Chen and Lin (1995) reported that oxygen consumption of *P. chinensis* juveniles (0.76±0.05 g) increased notably with pH level falling in the range of 7.0–

8.5 within 4 h. In addition, Wang et al. (2006) found that *L. vannamei* (0.76±0.05 g) were cultured in salinity (20–30) for 45 d, its growth energy and excretion energy changed markedly, but the rate of survival and growth had no significant difference; Dong et al. (1994) reported that energy consumption of *Macrobrachium nipponense* (1.380±0.385 g) obviously increased with pH rising (6.5→8.5) in 18 d, the ability of excretion and respiration enhanced, meanwhile growth rate were not affected by changing pH. Therefore, we consider that the environmental changes destroyed the osmoregulation equilibrium of the shrimp. The oxygen consumption and energy requirement were enhanced notably in short-time. When the alterant extent of environmental factors did not overstep the bounds of osmoregulatory scope, the survival rate did not change notably. However, the growth would be in normal or even slow level. Shrimp adjusted to environmental changes after a certain period, and finally, had no obvious difference in survival and growth in long term. This, we believe, is mostly related to the mechanism of physiological adaptation.

In conclusion, we considered that during the course of desalting and culturing shrimp in brackish water and freshwater regions, the key points lay in salinity and pH regulation, and a whole analysis of the osmoregulatory adaptation should be done, together with the survival and growth to ascertain a reasonable salinity and pH regulation criteria, which should have significant meaning to improve the healthy cultivation of shrimp. Base on this experiment, we might know that the activities of both total ATPase and special osmoregulatory undertakers showed significant variation in different salinity and pH. So, we suggested that during the process of desalting and culturing of *L. vannamei* postlarvae, the salinity changing should not exceed 3 and pH variety not more than 0.5.

Acknowledgement

The paper was supported by the Scientific Research Foundation for Outstanding Young Scientists of Shandong Province (2006BS07005), and the Sea-prosper Science and Technology Project of Shandong province of China (2001-3-6). The authors would like to thank all the peoples of Physiological Laboratory of Ocean University of China for assistance in the samples.

References

Allan, G.L., Maguire, G.B., 1992. Effects of pH and salinity on survival, growth and osmoregulation in (*Penaeus monodon*) Fabricius. Aquaculture 107, 33–47.

Bradford, M.M., 1976. A rapid and sensitive method for the quantitation of microgram quantities of protein utilizing the principle of protein dye binding. Anal. Biochem. 72, 248–254.

Breitwieser, G.E., Altamirano, A.A., Russell, J.M., 1987. Effects of pH changes on sodium pump fluxes in squid giant axon. Am. J. Physiol. 53 (C), 547–554.

Castille, F.L., Lawrence, A.L., 1981. The effect of salinity on the osmotic, sodium and chloride concentrations in the hemolymph of euryhaline shrimp of the genus *Penaeus*. Comp. Biochem. Physiol. A68, 75–80.

Charmantier, G., Soyez, C., 1994. Aquacop effect of molt stage and hypoxia on osmoregulatory capacity in the penaeid shrimp *Penaeus vannamei*. J. Exp. Mar. Biol. Ecol. 178, 233–246.

Chen, J.C., Chia, P.G., 1997. Osmotic and ionic concentrations of *Scylla serata* (Forskål) subjected to different salinity levels. Comp. Biochem. Physiol. 117A, 239–244.

Chen, J.C., Jun, L.L., 1994. Response of osmotic and chloride concentrations of *Penaeus chinensis* osbeck subadults acclimated to different salinity and temperature levels. J. Exp. Mar. Biol. Ecol. 179, 267–278.

Chen, J.C., Lin, C.Y., 1995. Responses of oxygen consumption, Ammonia-N excretion and Urea-N excretion of *Penaeus chinensis* exposed to ambient ammonia at different salinity and pH levels. Aquaculture 136, 243–255.

Chen, J.C., Liu, J.N., 1996. Survival, growth and intermolt period of juvenile *Penaeus chinensis* (Osbeck) reared at different combinations of salinity and temperature. J Exp Mar Bio Ecol 204, L169–L178.

Chen, J.C., Lin, Y.C., 2001. Acute toxicity of ammonia on *Litopenaeus vannamei* Boone juveniles at different salinity levels. J. Exp. Mar. Bio. and Eco. 259, 109–119.

Cheng, W., Chen, J.C., 1998. Penaeus chinensis juveniles reared at different salinity and temperature levels. Aquaculture 164, 173–181.

Christensen, J.D., Monaco, M.E., Lowery, T.A., 1997. An index to assess the sensitivity of Gulf of Mexico species to changes in estuarine salinity regimes. Gulf Res Rep 9 (4), 219–229.

Corotto, F.S., Holliday, C.W., 1996. Branchial Na^+–K^+-ATPase and osmoregulation in the purple shore crab *Hemigrapsus nudus* (Dana). Comp. Biochem. Physiol. 113A, 361–368.

Davis, D.A., Saoud, I.P., McGraw, W.J., Rouse, D.B., 2002. Considerations for *Litopenaeus vannamei* reared in inland low salinity waters. In: Cruz-Suarez, E., Rique-Marie, D., Tapia-Salazar, M., Gaxiola, G., Simoes, N. (Eds.), Avances en Nutrición Acuícola: Memorias del VI Simposium Internacional de Nutrición Acuícola, pp. 73–94. 3 al 6 de septiembre del.

Dong, S.L., Du, N.S., Lai, W., 1994. Effects of pH and Ca^{2+} concentration on growth and energy budget of *Macrobrachium nipponense*. J. Fish Chin. 18 (2), 118–128.

Ferraris, R.P., Parado-Estepa, F.D., Ladfa, J.M., 1986. Effect of salinity on the osmotic, loride, total protein and calcium concentrations in the hemolymph of the prawn *Penaeus monodon*. Comp. Biochem. Physiol. 83A, 701–708.

Forgac, M., 1998. Structure, function and regulation of the vacuolar (H+)-ATPases. FEBS 440, 258–263.

Furriel, R.P.M., McNamara, J.C., Leone, F.A., 2000. Characterization of Na^+–K^+-ATPase in gill microsomes of the freshwater shrimp *Macrobrachium olfersii*. Comp. Biochem. Physiol. 26B, 303–315.

Gerencser, G.A., Zhang, J.L., 2001. Cl^--ATPases: biological active transporters. Comp. Biochem. Physiol. 130A, 511–519.

Ponce-Palfox, J., Martinez-Palacios, C.A., Ross, L.G., 1997. The effects of salinity and temperature on the growth and survival rates of juvenile white shrimp, *Penaeus vannamei*, Boone, 1931. Aquaculture 157, 107–115.

Kirschner, L.B., 2004. The mechanism of sodium chloride uptake in hyperregulating aquatic animals. J. Exp. Biol. 207, 1439–1452.

Kosilo, B., Bigalke, T., Graszyski, K., 1988. Purification and characterization of gill (Na+–K+)-ATPase in the freshwater crayfish *Orconectes limosus* rutinesque. Comp. Biochem. Physiol. 89B, 171–177.

Kumlu, M., Eroldogan, O.T., 2000. Effects of temperature and salinity on larval growth survival and development of penaeus *semisulcatus*. Aquaculture 188, 167–173.

Lee, S.H., 1982. Low salinity adaptation of HCO_3^--dependent ATPase activity in the gills of blue crab (*Callinectes sapidus*). Biol Biophy Acta 689 (1), 143–154.

Lima, A.G., McNamara, J.C., Terra, W.R., 1997. Regulation of hemolymph osmolytes and gill Na^+/K^+-ATPase activities during acclimation to saline in the freshwater shrimp *Macrobrachium olfersii*(Wegmann,1836) (Decapoda, Palamonidae). J. Exp. Mar. Biol. Ecol. 215, 81–91.

López Mañanes, A.A., Meligeni, C.D., Goldemberg, A.L., 2002. Response to environmental salinity of Na^+–K^+-ATPase activity in individual gills of the euryhaline crab *Cyrtograpsus angulatus*. J. Exp Mar Biol Ecol 293, 320–335.

Lucu, Č., Flik, G., 1999. Na^+–K^+-ATPase and Na^+/Ca^{2+}exchange activities in gills of hyperregulating Carcinus maenas. Am. J. Physiol. 276, R490–R499.

Lucu, Č., Towle, D.W., 2003. Na^+–K^+-ATPase in gills of aquatic crustacea. Comp. Biochem. Physiol. 135 (A), 195–214.

Morris, S., 2001. Neuroendocrine regulation of osmoregulation and the evolution of air breathing in decapod crustaceans. J. Exp. Biol. 204 (5), 979–989.

Neufeld, G.J., Holliday, C.W., Pritchard, J.B., 1980. Salinity adaptation of gill Na^+–K^+-ATPase in the blue Crab, *Calliectes sapidus*. J Exp Zool 211, 215–224.

Oliveira, P.F., Lopes, I.A., Barrias, C., Rebelo da Costa, A.M., 2004. H^+-ATPase of crude homogenate of the outer mantle epithelium of *Anodonta cygnea*. Comp. Biochem. Physiol. 139, 425–432.

Onken, H., Putzenlechner, M., 1995. A V-ATPase drives active, electrogenic and Na^+-independent Cl^- absorption across the gills of *Eriocheir sinensis*. J. Exp. Biol. 198, 767–774.

Onken, H., Riestenpatt, S., 1998. NaCl absorption across split gill lamellae of hyperregulating crabs: transport mechanisms and their regulation. Comp. Biochem. Physiol. 119A, 883–893.

Pan, L.Q., Luan, Z.H., Jin, C.X., 2006. Effects of Na^+/K^+ and Mg^{2+}/Ca^{2+} ratios in saline groundwaters on Na^+–K^+-ATPase activity, survival and growth of *Marsupenaeus japonicus* postlarvae. Aquaculture 261, 1396–1402.

Péqueux, A., 1995. Osmotic regulation in crustaceans. J. Crustac. Biol. 15, 1–60.

Piller, S.C., Henry, R.P., Doeller, J.E., Kraus, D.W., 1995. A comparison of the gill physiology of two euryhaline crab species, *Callinectes sapidus* and *Callinectes similis*: energy production, transport-related enzymes and osmoregulation as a function of acclimation salinity. J. Exp. Biol. 198, 349–358.

Rosas, C., Sanchez, A., 1996. Dietary protein level on apparent heat increment and post-prandial nitrogen excretion of *Penaeus setiferus, P.schimitti, P.dnoraun and P.notialis* postlarvae. J. World Aquac. Soc. 27 (1), 92–102.

Saoud, I.P., Davis, D.A., Rouse, D.B., 2003. Suitability studies of inland well waters for *Litopenaeus vannamei* culture. Aquaculture 217, 373–383.

Shaw, J., 1960. The absorption of sodium ions by the crayfish *Astacus pallipes* Lereboullet.▯. The effect of other cations in the external solution. J. Exp. Biol. 37, 548–556.

Siebers, D., Leweck, K., Markus, H., Winkler, A., 1982. Sodium regulation in the shore crab *Carcinus maenas* as related to ambient salinity. Mar Biol 69, 37–43.

Stem, S., Bacat, A., Cahem, D., 1984. Characterization of Na^+, K^+-ATPase from the gills of the freshwater prawn *Macrobrachian rosenbergii*(De Man). Comp. Biochem. Physiol. 79B, 47–50.

Towle, D.W., 1981. Role of Na^+–K^+-ATPase in ionic regulation by marine and estuarine animals. Mar Biol Lett 2, 107–122.

Towle, D.W., Paulsen, R.S., Weihrauch, D., Kordylewski, M., Salvador, C., Lignot, J.H., Spaning-Pierrot, C., 2001. Na^+–K^+-ATPase in gills of the blue crab *Callinectes sapidus*: cDNA sequencing and salinity-related expression of a-subunit mRNA and protein. J. Exp. Biol. 204, 4005–4012.

Wang, W.N., Wang, A.L., Chen, L., Liu, Y., Sun, R.Y., 2002. Effects of pH on survival, phosphorus concentration, adenylate energy charge and Na^+–K^+-ATPase activities of *Penaeus chinensis* Osbeck juveniles. Aquat. Toxicol. 60, 75–83.

Wang, Xingqiang, Cao, Mei, Ma, Shen, Shuang-lin, Dong, Bin-lun, Yan, 2006. Effects of salinity on survival, growth and energy budget of juvenile *Litopenaeus vannamei*. Mar. Fish Res. 27 (1), 8–13.

Weihrauch, D., Ziegler, A., Siebers, D., Towle, D.W., 2002. Active ammonia excretion across the gills of the green shore crab *Carcinus maenas*: participation of Na^+/K^+-ATPase, V-type H^+-ATPase and functional microtubules. J. Exp. Biol. 205, 2765–2775.

Wilder, M.N., Ikuta, K., Atmomarsono, M., Hatta, T., Komuro, K., 1998. Changes in osmotic and ionic concentrations in the hemolymph of *Macrobrachium rosenbergii* exposed to varying salinities and correlation to ionic and crystalline composition of the cuticle. Comp. Biochem. Physiol. 119A, 941–950.

Zare, S., Greenaway, P., 1998. The effect of moulting and sodium depletion on sodium transport and the activities of Na^+–K^+-ATPase and V-ATPase in the freshwater crayfish C*herax destructor*. Comp. Biochem. Physiol. 119A (3), 739–745.